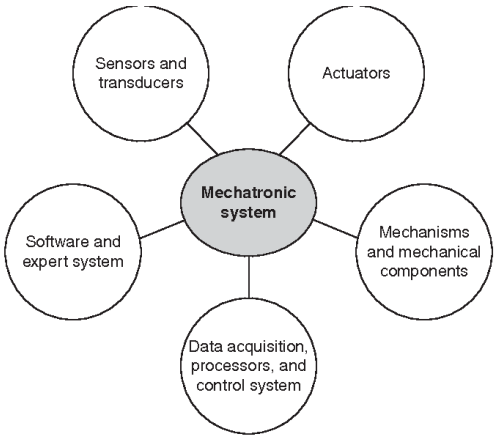
Mechatronics

"Synergistic integration of mechanical engineering with electronics and intelligent computer control in the design and manufacturing of industrial products and processes."



System

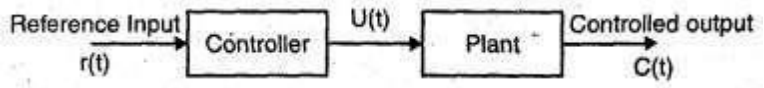
A system is an arrangement of or a combination of different physical components connected or related in such a manner so as to form an entire unit to attain a certain objective

Types of control systems

1) Open loop control system

2) Closed loop control system

Open loop control system: It is a system in which control action is independent of output. To each reference input there is a corresponding output which depends upon the system and its operating conditions.



Example: Electric heater without thermostat**,** traffic signals

Advantages of open loop system:

1. They are simple in construction and design.
2. They are economic.
3. Easy for maintenance.
4. Not much problem of stability.
5. Convenient to use when output is difficult to measure.

Disadvantages of open loop system:

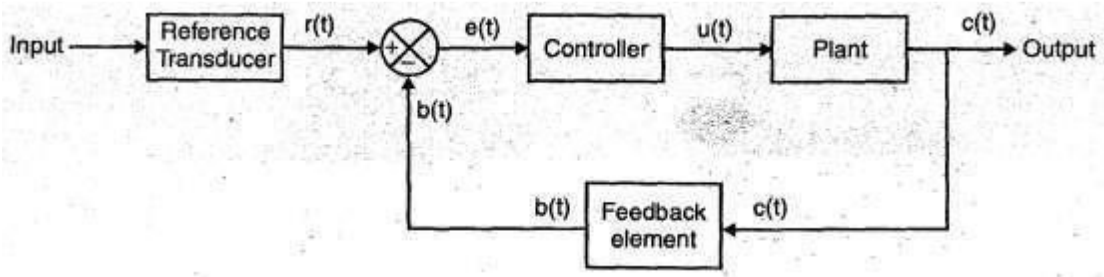
1. Inaccurate and unreliable because accuracy is dependent on accuracy of calibration.

2. Inaccurate results are obtained with parameter variations, internal disturbances.

3. To maintain quality and accuracy, recalibration of controller is necessary from time to time

Closed loop control system: Is a system in which the control action depends on the output.

A closed-loop control system, also known as a feedback control system, is a system that uses feedback to automatically control a process. It works by continuously monitoring the output, comparing it to the desired input, and adjusting the process to minimize any differences or errors. This feedback loop allows for self-correction, adaptation, and maintaining stability without manual intervention.



Example: Room Air conditioner, Automatic electric iron, water level controller

The key components:

- **Input:** The desired state or setpoint of the system (e.g., desired temperature).
- **Controller:** Receives the input and the feedback signal, calculates the necessary adjustments, and sends control signals to the process.
- **Process:** The system or mechanism being controlled (e.g., a heating system).
- **Sensor**: Measures the actual output of the process and provides feedback to the controller.
- **Feedback:** The signal representing the actual output, which is compared to the input to determine the error.

How it works:

1. The controller receives the desired input and compares it to the feedback signal from the sensor.
2. If there's a difference (error), the controller generates a control signal to adjust the process.

3. The process then changes its output based on the control signal.
4. The sensor continuously monitors the new output and provides updated feedback to the controller, forming a continuous feedback loop.
5. This loop continues until the actual output matches the desired input, minimizing the error.

Examples:

- Thermostat-controlled heating/cooling system: The thermostat acts as the controller, monitoring the room temperature (output) and adjusting the heating or cooling system (process) to maintain the desired temperature (input).
- Automatic electric iron: The temperature sensor monitors the heating element's temperature and adjusts the heating element to maintain the desired temperature.
- Water level controller: The level sensor monitors the water level and adjusts the water flow to maintain the desired level.

Advantages of closed loop system:

1. Accuracy is very high as any error arising is corrected.
2. It senses changes -in output due to environmental or parametric change, internal disturbance etc. and corrects the same.
3. High bandwidth.
4. Facilitates automation

Disadvantages

1. Complicated in design and maintenance costlier.
2. System may become unstable

Applications

1. Automatic Electric Iron
2. Servo Voltage Stabilizer
3. Water Level Controller
4. Cooling System in Car
5. Air Conditioner
6. Refrigerator
7. Missile Launched and Auto Tracked by Radar

Automation in industry

Automation refers to the use of technology to perform tasks or processes without human intervention. It involves the use of machines, software and systems to control and manage a range of functions. Automation can help increase efficiency, accuracy, and productivity, reducing costs and minimizing errors.

Types of Automation Fixed Automation

1. **Fixed automation**: also known as hard automation, is a type of automation system that is designed to perform a specific task or function, typically in a manufacturing or production environment. Fixed automation systems are highly specialized and often used for high-volume, repetitive tasks that require a high degree of precision

Industrial examples of fixed automation include:

- ➢ Assembly line in automobile manufacturing
- ➢ Production line in food processing
- ➢ Automated production line in electronics manufacturing
- ➢ Chemical processing plant

2. Programmable Automation: Programmable automation, also known as soft automation, is a type of automation system that can be programmed or reprogrammed to perform different tasks or functions.

Programmable automation systems are more flexible and adaptable than fixed automation systems, making them well-suited to applications that require frequent changes or adjustments.

These systems typically involve the use of computers and software to control and manage the automation process.

Examples of programmable automation systems:

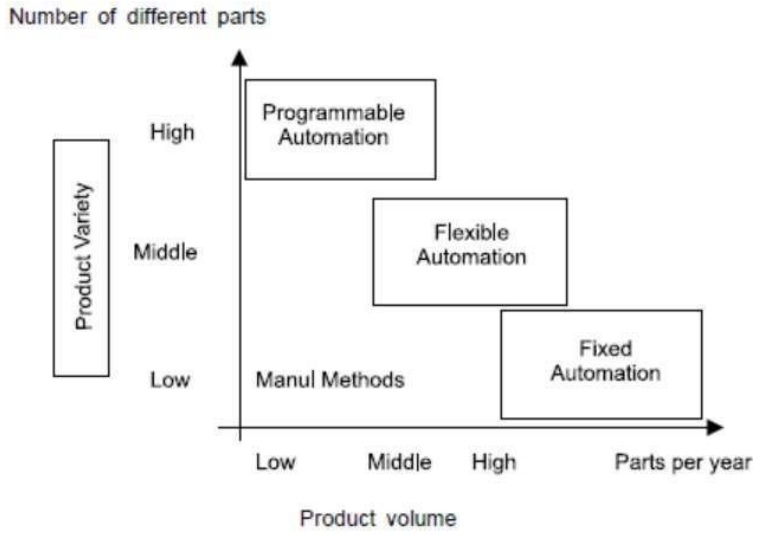
CNC machines in the machining industry

Automated assembly line in electronics manufacturing

Automated warehouse in the logistics industry

Advantages of programmable automation: is its versatility and adaptability, which can make it more cost-effective and efficient for smaller-scale or custom production runs.

Disadvantages: However, it can also be more complex and require more specialized skills to set up and operate than fixed automation systems.



3. Flexible Manufacturing : A Flexible Manufacturing System (FMS) is a production system designed to handle a variety of production needs with adaptability and efficiency. It uses automation and interconnected machines to quickly adjust to changes in product design, production volume, or even the order of operations. This adaptability allows manufacturers to respond to market demands, including customized products, with greater speed and efficiency compared to traditional mass production systems.

Flexible automation systems typically involve the use of robotics, computer controls, and other advanced technologies to enable machines to perform a wide range of tasks, such as sorting, assembly, and packaging.

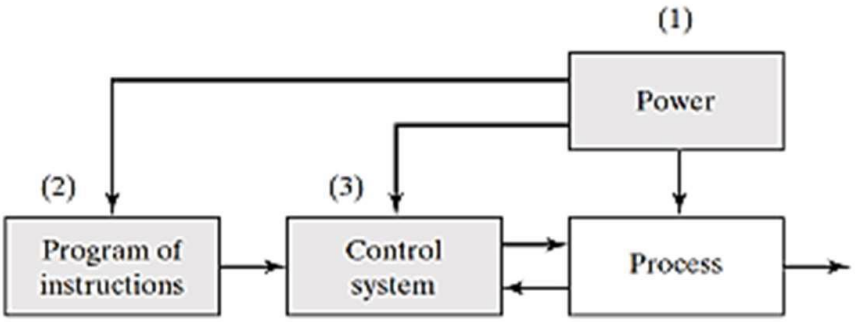
Examples of FMS Applications:

Aerospace industry, Automotive industry, and Electronics manufacturing.

Basic Elements of Automation

- **Automated Machines:** CNC machines, robots, and other programmable devices.
- **Material Handling Systems:** Automated storage and retrieval systems (AS/RS), conveyors, and automated guided vehicles (AGVs).
- **Control Systems:** Centralized computer system for managing the entire manufacturing process.
- **Human-Machine Interfaces:** Enables operators to interact with and monitor the system.
- **Sensing Devices**: These are devices that can detect changes in the environment, such as temperature, pressure, or motion, and convert them into electrical signals that can be processed by a computer or control system.
- **Control Devices**: These are devices that receive input from the sensing devices and use that information to control the operation of machines or equipment, such as motors, valves, or actuators.
- **Actuators**: These are devices that convert electrical signals from the control system into physical motion or force, such as hydraulic or pneumatic cylinders, or electric motors.

- **Power Source**: This is the energy source that provides power to the machines or equipment being controlled, such as electricity or compressed air.
- Control System: This is the software or hardware that controls the operation of the machines or equipment, and processes data from the sensing devices and control devices to make decisions about how to control the system.



Advantages of Automation

Increased Efficiency: Automation can perform tasks faster and more consistently than humans, which can lead to increased productivity and efficiency in manufacturing and other industries.

Improved Quality: Automation can help reduce errors and defects by ensuring that tasks are performed consistently and accurately.

Reduced Labour Costs: Automation can help reduce labour costs by replacing some or all the human workers required to perform a task.

Increased Safety: Automation can help reduce the risk of workplace accidents and injuries by eliminating the need for workers to perform dangerous tasks.

Increased Flexibility: Automation can help increase flexibility in manufacturing and other industries by allowing tasks to be reprogrammed or reconfigured quickly and easily.

Improved Data Collection and Analysis: Automation can provide real-time data on the performance of machines and equipment, which can be used to improve efficiency, reduce downtime, and optimize production processes.

Disadvantages of Automation

While automation can provide many benefits, there are also some disadvantages to consider:

High Initial Investment Costs: Setting up an automated system can require a significant initial investment in equipment, software, and training.

Potential Job Loss: Automation can lead to the displacement of human workers, particularly in industries with high levels of repetitive or routine tasks.

Complexity And Maintenance Requirements: Automated systems can be complex and require specialized skills and knowledge to design, program, and maintain.

Lack Of Flexibility: Automated systems may not be as flexible as human workers, particularly in situations where tasks require adaptation to changing conditions.

Dependence on Technology: Automated systems may be vulnerable to technological malfunctions or failures, which can cause disruptions in production and require costly repairs.

Reduced Creativity and Problem-Solving: Automation can limit the opportunities for human workers to use their creativity and problem-solving skills in the workplace.

Introduction to Internet of Things

The Internet of Things (IoT) describes the network of physical objects—"things"—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.

Examples in Different IoT Applications:

- **Smart Agriculture:**

Sensors (soil moisture, temperature), microcontrollers, and communication modules (LoRaWAN) to monitor crops and environmental conditions.

- **Smart Homes:**

Sensors (motion, temperature, light), actuators (lights, thermostats), and communication modules (Wi-Fi, Bluetooth) to automate and control various aspects of the home.

- **Industrial IoT (IIoT):**

Sensors (pressure, flow, vibration), actuators (valves, motors), and communication modules (cellular, Ethernet) to monitor and control industrial processes.

Characteristics of IoT:

Intelligence: IoT systems are extensively liked in the market because of their intelligence. A combination of algorithms and computer enables the system to inform change in the environment and take appropriate actions. For example – Systems are intelligent enough to sense a sudden spike in temperature and trigger an alarm for fire.

Connectivity: Connectivity is the main characteristic of IoT as it enables the system to send data and stay connected to other devices. It provides system network accessibility and function collaboratively.

Expressing: IoT is all about interacting intelligently with the outer environment and humans. Expressing enables this interactivity. Expressing allows us to show output into the real world and input from people and the environment.

Sensing: Sensitivity means aware of the changes around us. Sensor technologies provide us with the means to create an experience that reflects an awareness of changes in the physical world and the people in it. It helps in expressing.

This forms the input for the IoT system and provides a better understanding of the complex world around us.

Energy: Everything in this world is driven by energy. IoT systems are created smart enough to synthesize energy from the outer environment and conserve it. It is also made energy efficient to work for a longer duration.

Security: Safety and security is the most important feature of any system. If the system is not secure to cyber-attack and illegal intervention, nobody will use it. IoT systems deal with personal data; that's why it's an obligation that all safety measure

should be taken care of in this system. All IoT systems are secure enough to deal with personal data.

Advantages:

- It can assist in the smarter control of homes and cities via mobile phones. It enhances security and offers personal protection.
- By automating activities, it saves us a lot of time.
- Information is easily accessible, even if we are far away from our actual location, and it is updated frequently in real time.
- Electric Devices are directly connected and communicate with a controller computer, such as a cell phone, resulting in efficient electricity use. As a result, there will be no unnecessary use of electricity equipment.
- Personal assistance can be provided by IoT apps, which can alert you to your regular plans.
- It is useful for safety because it senses any potential danger and warns users. For example, GM OnStar, is a integrated device that system which identifies a car crash or accident on road. It immediately makes a call if an accident or crash is found.
- It minimizes human effort because IoT devices connect and communicate with one another and perform a variety of tasks without the need for human intervention.
- Patient care can be performed more effectively in real time without the need for a doctor's visit. It gives them the ability to make choices as well as provide evidence-based care.
- Asset tracking, traffic or transportation tracking, inventory control, delivery, surveillance, individual order tracking, and customer management can all be made more cost-effective with the right tracking system.

Disadvantages:

- Hackers may gain access to the system and steal personal information. Since we add so many devices to the internet, there is a risk that our information as it can be misused.
- They rely heavily on the internet and are unable to function effectively without it.
- With the complexity of systems, there are many ways for them to fail.
- We lose control of our lives—our lives will be fully controlled and reliant on technology.
- Overuse of the Internet and technology makes people unintelligent because they rely on smart devices instead of doing physical work, causing them to become lazy.
- Unskilled workers are at a high risk of losing their jobs, which could lead to unemployment. Smart surveillance cameras, robots, smart ironing systems,

smart washing machines, and other facilities are replacing security guards, maids, ironmen, and dry-cleaning services etc.

- It is very difficult to plan, build, manage, and enable a broad technology to IoT framework.
- Deploying IoT devices is very costly and time-consuming.

The physical design of IoT (Internet of Things): refers to the tangible components that make up an IoT system, including sensors, actuators, microcontrollers, communication modules, and power supplies.

Key Components of the Physical Design:

- **Sensors:** These devices collect data from the physical environment, such as temperature, pressure, light, or motion.
- **Actuators:** Actuators are responsible for taking actions based on the data received, like controlling motors, lights, or valves.
- **Microcontrollers/Embedded Systems:** These are the "brains" of the IoT devices, processing data from sensors and controlling actuators.
- **Communication Modules:** These modules enable the devices to connect to a network, using technologies like Wi-Fi, Bluetooth, or cellular networks.
- **Power Supplies:** Providing the necessary energy for the devices to operate.
- **Gateways:** These act as bridges between the local network of IoT devices and the internet, often performing tasks like data aggregation and pre-processing.

Protocols in Physical Design:

- **Link Layer Protocols:** These protocols manage the physical connection between devices, like Ethernet (802.3) and Wi-Fi (802.11).
- **Network/Internet Layer Protocols:** These protocols handle routing and addressing of data packets, like IPv4 and IPv6.
- **Transport Layer Protocols:** These protocols manage the reliable transmission of data, like TCP and UDP.
- **Application Layer Protocols:** These protocols define how applications interact with the network, such as HTTP, CoAP, and MQTT.

The logical design of an IoT system: This focuses on how the system functions and interacts, without delving into the specific hardware or implementation details. It's an abstract blueprint that outlines the system's behavior, data flow, and how its components connect and communicate. Think of it as a map showing the relationships and interactions between devices, cloud platforms, and applications.

Key aspects of the logical design include:

- **Functional Blocks:**

These are the core components of the IoT system, providing capabilities like identification, sensing, actuation, communication, and management.

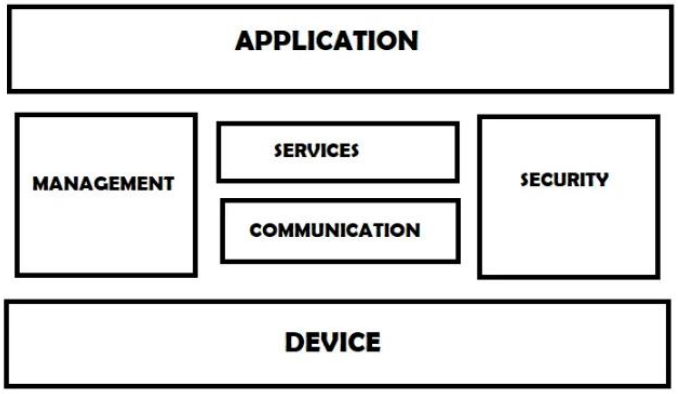
- **Communication Models:**

These define how data is exchanged between different parts of the system. Common models include request-response, publish-subscribe, and push-pull.

- **Communication APIs:**

These are the interfaces used for communication between devices and other components of the system, such as REST APIs or WebSocket connections.

In essence, the logical design answers the question of how the IoT system will achieve its intended functionality, rather than focusing on the specific physical components or their implementation.



Here's a breakdown of the key elements:

- **Device Identification:** Defining how each device in the IoT network will be uniquely identified.
- **Data Sources:** Specifying where the data originates from within the system.
- **Network Topology:** Outlining how devices are connected and how they communicate with each other.
- **Service Groups:** Defining logical groupings of devices or functionalities.
- **Control Structures:** Establishing how the system will be controlled and managed.

Benefits of a well-defined logical design:

- **Abstraction:** Allows for simulating the IoT environment before deploying physical devices.
- **Flexibility:** Enables modifications and updates to the system without affecting the underlying hardware.

- **Scalability:** Helps in designing a system that can handle a growing number of devices and data volume.
- **Interoperability:** Ensures that different devices and systems can communicate and work together seamlessly.
- **Security:** Allows for incorporating security measures like authentication, authorization, and data encryption.

IoT protocols are sets of rules and standards that enable devices to communicate and exchange data within an Internet of Things (IoT) network. They define how devices discover each other, connect, transmit data, and ensure security. These protocols are crucial for the successful operation of IoT systems, allowing different devices, regardless of manufacturer or platform, to interact and exchange information.

Here's a more detailed look at IoT protocols:

1. Purpose of IoT Protocols:

- **Interoperability:** IoT protocols ensure that different devices, even those from different vendors, can communicate with each other.
- **Standardization:** They provide a common language for devices to understand and process data.
- **Efficiency:** IoT protocols are designed to be lightweight and efficient, suitable for devices with limited resources.
- **Security:** They incorporate security measures to protect data transmitted between devices.

2. Key Categories of IoT Protocols:

- **Networking Protocols:** These protocols define how devices connect to networks and transmit data. Examples include:

  Wi-Fi: Widely used for local area networks.

  Bluetooth/BLE: Commonly used for short-range communication, especially in wearables and smart home devices.

  Zigbee: A mesh network protocol for low-power, short-range connections, popular in home automation.

  Cellular IoT (2G, 3G, 4G, 5G): Used for connecting devices over cellular networks.

  Thread: A low-power, IP-based mesh networking protocol designed for IoT.

3. **Communication Protocols:**

These protocols handle the actual exchange of data between devices. Examples include:

MQTT (Message Queuing Telemetry Transport): A lightweight messaging protocol ideal for IoT devices with limited resources.

CoAP (Constrained Application Protocol): A specialized protocol for constrained devices and networks.

AMQP (Advanced Message Queuing Protocol): A more robust messaging protocol often used in enterprise environments.

HTTP (Hypertext Transfer Protocol): The foundation of the web, also used in some IoT applications.

XMPP (Extensible Messaging and Presence Protocol): Used for real-time communication, including messaging and presence information.

DDS (Data Distribution Service): A data-centric middleware for real-time systems, offering efficient communication between devices.

4. **Data Protocols:**

These protocols manage the format and structure of data being exchanged.

5. **Gateway Protocols:**

These protocols enable communication between IoT devices and cloud platforms. Examples include MQTT, HTTP, CoAP, AMQP, and WebSocket.

Choosing the Right Protocol:

The selection of an IoT protocol depends on various factors, including:

- **Range and Bandwidth Requirements:** Short-range, low-power devices might use Bluetooth or Zigbee, while devices needing wider coverage could use cellular or LoRaWAN.
- **Power Consumption:** Battery-powered devices require protocols with low power consumption, like Bluetooth Low Energy (BLE) or Zigbee.
- **Security Needs:** Protocols like MQTT offer security features like encryption and authentication.
- **Scalability:** The protocol should be able to handle the number of devices and data volume expected in the application.
- **Complexity and Cost:** Some protocols are simpler to implement and deploy than others, impacting development time and cost.

The robotics configuration:

Cartesian Configuration Robots:

▪ Provides three linear motions along three mutually perpendicular axes: X, Y, and Z. However, there is no rotary motion.

▪ Configuration provides rectangular work envelope.

▪ Used for assembly, palletizing and machine tool loading.

Advantages: 1. Simple controls due to linear movements and easy to visualize

2. High degree of mechanical rigidity, accuracy and repeatability

3. Can carry heavy loads

4. Motion of the wrist end is smoother

Disadvantages:

1. The arm movement is limited to a small rectangular work space

2. It occupies large area there by giving a low ratio robot size to operating volume

Applications: Assembly, welding, machine loading/unloading, surface finishing, inspection, etc.

Cylindrical Configuration Robots:

▪ Provides two linear and one rotary motion.

▪ Configuration provides cylindrical work envelope, has good work area to floor area ratio.

▪ Used for loading and unloading on machine tools.

Advantages: 1. Larger work space than Cartesian configuration 2. Robot is relatively easy to program

Disadvantages: 1. Overall mechanical rigidity is reduced because robots with a rotary axis must overcome the inertia of the object when rotating

2. Robots repeatability and accuracy is also reduced in the direction of rotary movement

3. It requires more sophisticated control system

Applications: Pick and place operations, machine loading/unloading, investment casting, forging operations, conveyer pellet transfers, assembly, coating applications, etc.

Polar (Spherical) Configuration Robots:

▪ Provides one linear and two rotary motions.

▪ Configuration provides spherical work envelope.

▪ Used for spot welding and manipulation (Handling) of heavy loads

Advantages: 1. It can generate large working envelope compared to Cartesian and cylindrical

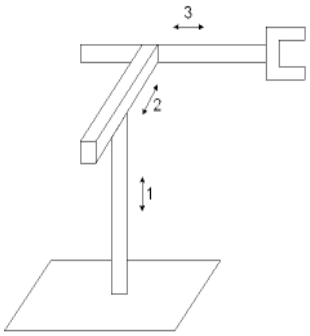
2. It is simple in design and easy to program

3. Provides good weight lifting capabilities

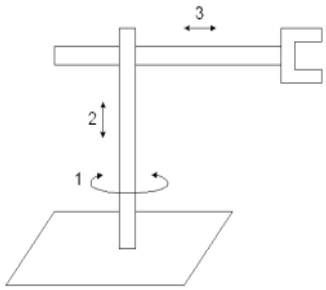
Disadvantages:

1. Reduced mechanical rigidity
2. More sophisticated control system is required
3. Limited vertical movement

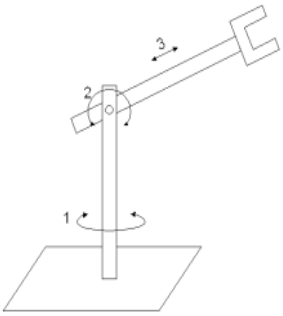
Applications: Die casting, injection moulding, forging, machine tool press loading/unloading, dip coating, material transfer applications, etc.



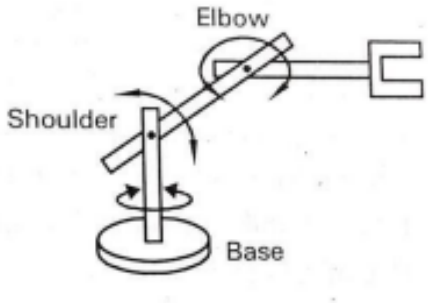
Cartesian: LOO



Cylindrical: TLO



Spherical (Polar): TRL



Jointed Arm Configuration Robot

Jointed Arm Configuration Robot: It is like that of human arm, consisting of two straight components corresponding to the human forearm and upper arm mounted on a vertical pedestal that can be rotated about the base so as to provide the robot with the capacity to work within a quasi-spherical space. It gives rise to three rotational movements.

Advantages:

1. Most flexible configuration that enables the robot arm to move in all directions
2. Large working envelope
3. Versatile configuration

Disadvantages:

1. Requires complex programming
2. Control actions are different
3. Less stable as the arm approaches maximum reach

Applications: Painting, welding, automatic assembly, welding, pick and place operations involving light parts, etc